P1

(i) $Y \sim Exp(\mu)$ $X_i \sim Exp(\theta)$

$$\min\{Y, X_1, \dots X_n\} \sim Exp(\mu+n\theta)$$

$$E[\min\{Y, X_1, \dots X_n\}^{-1/2}] = \int_0^\infty x^{-1/2} \frac{1}{\mu+n\theta} e^{-\frac{x}{\mu+n\theta}} dx$$

$$= \sqrt{\pi(\mu+n\theta)}$$

(ii) set $W = \max\{Y, X_1 \dots X_n\}$

$$P(W \le w) = P(Y \le w) \cdot P(X_i \le w, \forall i = 1, \dots, n)$$

$$= (1-e^{-\mu w})(1-e^{-\theta w})^n$$

$$E[W] = \int_0^\infty w f_W(w) dw = \int_0^\infty P(W > w) dw = \int_0^\infty (1 - P(W \le w)) dw$$

$$= \int_0^\infty [1 - (1-e^{-\mu w})(1-e^{-\theta w})^n] dw$$

P_2

set T_i is the waiting time of i-th customer behind Eaton

C is the service time of Eaton

$$P(T_i < C \ \forall i = 1, \dots, n) = \frac{\theta^n}{(\theta+\mu)^n}$$

P_3

(a) $P = 0$

(b) set T_A, T_B, T_C, T_d are the service time of ABCD respectively.

$$P(T_A > T_C + T_d, T_B > T_C + T_d)$$

$$= P(T_d < T_A - T_C, T_d < T_B - T_C, T_A > T_C, T_B > T_C)$$

$$= P(T_d < T_A - T_C \mid T_A > T_C) P(T_A > T_C) P(T_d < T_B - T_C \mid T_B > T_C) P(T_B > T_C)$$

$$= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{16}$$

P4

(a) let $N(t)$ denote the # of waiting customer

$N(t)$ is a P.P. with $\lambda = 7$

set T is the time train arrive . $T \sim Uni(0,1)$

we need to get $N(T)$.

$X = N(T) \sim Poisson(\lambda T)$

$$E[N(T)] = E[E[N(T)|T]] = \frac{7}{2}$$

(b) $Var(X) = Var(E[X|T]) + E[Var(X|T)]$

$$= Var(7T) + E[7T] = \frac{49}{12} + \frac{7}{2} = \frac{91}{12}$$

P5

(a) set $N(t)$ is # of car pass the intersection by time t

$N_i(t)$ is # of type i car . . . .

$N_i(t) \sim Poisson(\lambda t p_i)$

$$P(N_1(t) \geq 1 , N_2(t) \geq 1 , N_3(t) \geq 1 , N_4(t) = 0)$$

$$= (1 - e^{-\lambda t p_1})(1 - e^{-\lambda t p_2})(1 - e^{-\lambda t p_3})(e^{-\lambda t p_4})$$

(b) $P(N_1(t) = 4 \mid N_1(t) + N_2(t) = 6)$

$$= \frac{\frac{(\lambda p_1 t)^4}{4!} e^{-\lambda p_1 t} \frac{(\lambda p_2 t)^2}{2!} e^{-\lambda p_2 t}}{\frac{[\lambda(p_1+p_2)t]^6}{6} e^{-\lambda(p_1+p_2)t}}$$